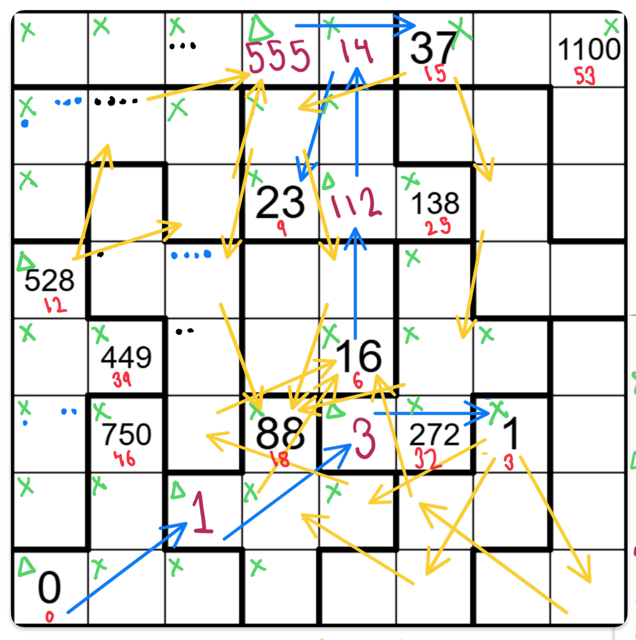
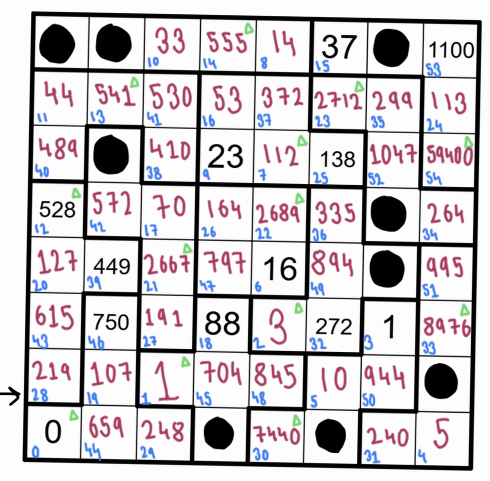


• 1st level Move
• 2nd "   "
• 3rd "   "

x Tower not Possible
△ Tower at that Posn
• → Score
• → Path
• Possible Path

3 Possible Paths 1 → 16
4 ———— 23 → 528
2 ———— 528 → 555
3 ———— 37 → 88

Knight's Start →



●	●	33	555	14	37	●	1100
44	541	530	53	372	2712	299	113
489	●	410	23	112	138	1047	59400
528	572	70	164	2684	335	●	264
127	449	2667	797	16	894	●	995
615	750	191	88	3	272	1	8976
219	107	1	704	845	10	944	●
0	659	248	●	7440	●	240	5

The region above has been tiled with the 12 pentominoes (plus a 2-by-2 tetromino) into 13 regions. Think of each of these 13 regions as constructed out of 1-by-1-by-1 cubes. We need to add a **tower** to each region. A tower is an additional size-1 cube placed on one of a region's squares.

After adding these towers, place a knight at the bottom-left square. It then proceeds to make knight's moves until it has visited all the towers. It never visits the same space twice. (A *move* on this board involves travelling 0 units in one dimension, 1 in another, and 2 in the third. The knight is allowed to "pass through" towers as it moves.)

But there's a catch: As you can see, the knight starts with a score of 0. On its *N*th move, its score increases by *N* if the move is to a location at the same altitude as the square it moved from. If, instead, it moves up, the score is multiplied by *N*. And finally, if it moves down, the score is divided by *N*. This last type of move is only allowed if the score is evenly divisible by *N*.

Every three moves, up until move #18, the knight wrote down its score upon arriving at a given square. From then on it only wrote down its score every *K* moves, for some larger value *K*. Using this information, **can you reconstruct the knight's path?**

After filling all the remaining visited squares with the missing score values, find the *unvisited* squares. For each of these squares, compute the sum of the scores in any orthogonally adjacent squares that were part of the knight's path. The answer to this puzzle is the sum of these "neighbor sums" from the unvisited squares.

→ 5^12 x4 possibilities → 976,562,500

→ means max 64 −1 (as 0 already occupied) = 63 moves

A pentomino puzzle is a spatial and geometric puzzle based on "pentominoes"—shapes formed by connecting exactly five equal-sized squares edge-to-edge.

Done
→ Possible Posn of Tower (1) but necessary

1 coming at 1 at N_1^{th} Move
Let Previous Score n_1
α + same floor $n_1 = 1 - N_1$ ⇒ N_1 1 α
α × tower at 1 $n_1 \times N_1 = 1$ ⇒ $n_1 = \frac{1}{N_1} > 1$ ⇒ $N_1 = 1$ α
✓ ÷ ⇒ No tower at 1 $\frac{n_1}{N_1} = 1$ $\boxed{n_1 = N_1}$

2 coming at 16 at N_2^{th} Move
Previous Score n_2
+ same floor $n_2 = 16 - N_2$ ⇒ $N_2 \in \{3, 6, 9, 12, 15\}$
× tower at 16 $n_2 = \frac{16}{N_2}$ ⇒ $N_2 = [1, 2, 4, 8, 16]$ α α α α α as we write at interval of 3
÷ No tower at 16 $n_2 = 16 N_2$
No possible N_2 ⇒ No tower at 16 but can be from +

3 N_3^{th} Move
n_3
+ $n_3 = 23 - N_3$ ⇒ $N_3 \in \{3, 6, 9, 12, 15, 18, 22, 23\}$ (K=4, K=5)
× $n_3 = \frac{23}{N_3}$ ⇒ $N_3 \in \{23\}$
÷ $n_3 = 23 N_3$

4 N_4^{th} Move
n_4
+ $n_4 = 37 - N_4$ ⇒ $N_4 \in \{6-18, \ldots\}$ not 3 as minm move req 4
× $n_4 = \frac{37}{N_4}$ ⇒ $N_4 \in \{37\}$ ✗
÷ $n_4 = 37 N_4$

5 N_5^{th} Move
n_5 ÷, + No restriction from this point
× $n_5 = \frac{88}{N_5}$ ⇒ $N_5 \in \{22\}$ K=4

3, 6, 9, 12, 15, 18
6 moves → 6 scores written
left 57 "
Remaining scores written = 5
18+K, 18+2K, 18+3K, 18+4K, 18+5K
K > 3 & ∈ N

K=4	22	26	30	34	38
K=5	23	28	33	38	43
K=6	24	30	36	42	48
K=7	25	32	39	46	53
K=8	26	34	42	50	58
K=9	27	36	45	54	63

Distinct → 24
22 23 24 25 26^2 27 28 30^2
32 33 34^2 36^2 38^2 39 42^2 43
45 46 48 50 53 54 58 63

Next Move shouldn't be possible

	K		
×	4	38	→ 42 ✓
×	5	43	→ 48 ✓
×	6	48	→ 54 ✓
×	7	53	→ 60 ✓
✓	8	58	→ 66 ✗
✓	9	63	→ 73 ✗
	10	68 ✗	

∴ K = 8 or 9
✗ Wrong Conclusion it says until it visits all the towers not all squares So only valid conclusion K ∈ [4, 9]
∴ K might be 7 as Maybe Hinted in Puzzle's Name

[(528 _ 13) _ 14] _ 15 =
~~+ + + = 570~~ (+ + + = 570)
~~+ + × = 8325~~
+ + ÷ = 37 ✓
~~+ × + = 7589~~
α + × (× ÷)
α + ÷ (+ × ÷)
α × (+ × ÷) (+ × ÷)
α ÷ (+ × ÷) (+ × ÷)
only 37 via [+ + ÷]

[(37 _ 16) _ 17] _ 18 = △
+ + + = 88 ✓
+ + × = 1260
α + + ÷
+ × + = 919
α + × (×, ÷)
α + ÷ (+ × ÷)
× + + = 627
α × + (×, ÷)
α × × (+ × ÷)
α × ÷ (+ × ÷)
α ÷ (+ × ÷) (+ × ÷)
only 88 possible via + + +

Now from Score 1 the Maxm Possible Score at 6th Move → {16, 23, 37} ✓ × ×
Let's see Production/generation
[(1 _ 4) _ 5] _ 6 =
+ + + = 16 ✓
+ + × = 60
α + + ÷
+ × + = 31
α + × ×
α + × ÷
α + ÷ (+, × ÷)
× + + = 15
α × + ×
α × + ÷
α × × (+ × ÷)
α × ÷ (+ × ÷)
α ÷ (+ × ÷) (+ × ÷)
⇒ 16 is 6th Move Score & only possible combn for 16 is (+ + +)

Now from Score 16 the Maxm Possible Score → [(16 + 7) + 8] × 9 = 279
[(16 _ 7) _ 8] _ 9 =
+ + + = 40
+ + × = 279
α + + ÷
+ × + = 143
α + × ×
α + × ÷
α + ÷ (+ × ÷)
× + + = 129
α × + (× ÷)
α × × (+ × ÷)
× ÷ + = 23 ✓
× ÷ × = 126
α × ÷ ÷
α ÷ (+ × ÷) (+ × ÷)
⇒ 23 is the 9th Move Score & only possible combn for 23 is (× ÷ +)

Possible Score at 3rd Move
By minm move count → 1, 16, 23, × Wrong approach
Maxm Score generated by 3 Moves
[(0 + 1) + 2] × 3 = 9
only 1 possible to generate only possible moves
$[\underbrace{0 + 1 + 2}_{\text{1st Move, 2nd Move}}] \underbrace{/ 3}_{\text{3rd Move}} = 1$ △
Means already elevated

[(23 _ 10) _ 11] _ 12
+ + + = 56
+ + × = 528 ✓
α + + ÷
+ × + = 375
α + × (×, ÷)
~~+ ÷ + = 15~~
~~+ ÷ × = 36~~
α + ÷ ÷
× + + = 253
α × + (×, ÷)
α × × (+, × ÷)
α × ÷ (+ × ÷)
α ÷ (+ × ÷) (+ × ÷)
⇒ 528 is the 12th Move Score & only possible combn is (+ + ×)

88 K = 4 α (Algorithm)
K = 5 α "
K = 6 α "
$\boxed{K = 7}$ ✓ 138 " [+, +, ×, +, +, ÷, +] (8th tower, 9th, 10th)
K = 8 α "
K = 9 α "

138 → 272 [+ + + + × ÷ +] (11th)
272 → 449 [× ÷ + + + + +] (12th)
449 → 750 [+ + + + + + +]
750 → 1100 [+ + + + + + +]
7 towers visited up until move #18

Also 1100 (53rd Move) is not the last move as by virtue of existence of this still 1 more tower to reach as game ends when all " reached.
only $5^3 \times 2 \times 4 \times 3^2 = 9000$ possibilities left from almost 1 billion for the possible combn of tower posn.

Algorithm Ideas to find exact path to reach 1100 Score square (Backtracker)
⇒ 8×8 Height Matrix with 37 (6 towers) confirmed Heights (Maybe 0 for unknown, & 1 & 2)
⇒ we can totally leave out the score calculation (extra burden not required as they were there to just help find the exact combn of + × ÷)
⇒ One list containing 53 exact order of operations as we have found the exact order from which what type of move to take can be determined L or 2 in st line
⇒ Two types of possible moves L for + & ———— for × ÷
⇒ Should use co-ordinate System (1,1) to 0 Score & (8,8) to 1100 Score sq
⇒ Somehow define pentomino & tetromino otherwise no. of possible combn rise from 9000 to $^{27}C_7$ = 888,030 combn(s)
⇒ If any unknown height say has same height in all possible paths from a Node it should be updated in 8×8 update height matrix which should be discarded if all possible paths from the Node doesn't reach destination. This also means all future paths from that Node should use height + update height matrix (Update height Matrix initially have all 0 entries)